Literature Survey

Banking Policy Question Answering using RAG with Citations

1. Comparison of Reviewed Papers

#	Paper	Year	Problem it solved	Method used	Result obtained	Limitation
1	**Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks (Lewis et al.)**	2020	Language models keep knowledge inside their parameters, so it is hard to update, can be wrong, and comes with no source.	Combined a retriever (finds relevant Wikipedia passages) with a text generator that writes the answer using those passages. This is the original RAG idea.	Reached state-of-the-art results on several open-domain QA tasks and gave more factual answers than a generator alone.	Works on general Wikipedia knowledge, not banking. Does not show user-friendly citations.
2	**Enabling Large Language Models to Generate Text with Citations (ALCE) (Gao et al.)**	2023	LLM answers can be made up, and there was no standard automatic way to check whether the citations they give actually support the answer.	Built ALCE, a benchmark (ASQA, QAMPARI and ELI5 datasets) where a system must retrieve evidence and answer with citations. Introduced metrics for fluency, correctness and citation quality.	Even the best models often failed to fully support their answers; on ELI5 about half of the answers lacked complete citation support.	Uses general-knowledge questions, not banking. It is an evaluation benchmark, not a ready-to-use application.
3	**RIRAG: Regulatory Information Retrieval and Answer Generation (Gokhan et al. (RIRAG))**	2024	Regulatory documents are long, formal and cross-referenced, so finding the exact rules that answer a question is hard.	Created the ObliQA dataset (27,869 questions from 40 Abu Dhabi Global Markets financial regulation documents), defined a retrieve-then-answer task, and built a baseline RIRAG system.	Provided a large dataset and a working baseline that show regulatory question answering can be studied and measured.	Covers one regulator's documents, with questions generated automatically using GPT-4. Does not focus on page-level citations for end users.
4	**A Hybrid Approach to Information Retrieval and Answer Generation for Regulatory Texts (Hybrid regulatory IR paper)**	2025	Traditional keyword search struggles with the language of regulatory text.	Built a hybrid retrieval system that combines lexical (keyword) and semantic (meaning-based) search, then used RAG to generate answers. Tested on the ObliQA dataset.	Reported better retrieval from large regulatory collections and detailed answers that address compliance requirements.	Uses the same ADGM dataset. Focuses on retrieval and answer quality, not on a user-facing citation feature.
5	**A Novel Multi-Step Prompt Approach for LLM-based Q&As on Banking Supervisory Regulation (Licari et al.)**	2024	Compliance officers spend a lot of time answering questions about complex European banking supervisory reporting rules.	Designed a multi-step prompt method using GPT-4o and RAG, with an LLM acting as evaluator, on questions from the European Banking Authority Q&A.	Showed that LLMs can help answer such regulatory questions, while the authors stress that human verification is still needed.	Narrow topic (reporting obligations under EU rules). Not about Indian bank policies or customer-facing questions.
6	**FinRAG-12B: a production-validated recipe for grounded question answering in banking (FinRAG-12B)**	2026	Banks hesitate to adopt LLMs because of hallucination and the need for accurate, verifiable, grounded answers.	Trained a 12-billion-parameter model for banking RAG in two stages: general financial data first, then proprietary banking conversations and synthetic QA from SEC filings. The model must refuse to answer when the information is not available.	Presented as a production-validated recipe for grounded banking QA (see the paper for detailed numbers).	Depends on proprietary bank data and heavy model training, which students cannot easily reproduce.

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7	**FinSage: A Multi-aspect RAG System for Financial Filings Question Answering (FinSage)**	2025	Financial filings are long and multi-modal (text, tables, figures), which makes accurate question answering difficult.	Proposed a multi-aspect RAG framework tailored for regulatory compliance analysis of financial filings.	Reached 92.51% recall on 75 expert-curated questions and beat the best baseline on FinanceBench by 24.06% in accuracy. It was deployed and served more than 1,200 people.	Targets company filings and compliance analysis, with a complex design. Not aimed at customer-facing bank policy questions.
8	**Financial Report Chunking for Effective Retrieval Augmented Generation (Jimeno Yepes et al.)**	2024	How a document is split into chunks strongly affects RAG quality, and fixed-size chunks can break context.	Proposed element-based chunking that follows document structure (titles, paragraphs, tables) and compared it with token-size chunking on the FinanceBench dataset using SEC reports.	Element-based chunking gave better retrieval accuracy (reported up to 84.4%) and better Q&A results than token-based chunking.	Studied US SEC financial reports only. Does not deal with citations or bank policy documents.

2. Short Introduction to Each Paper

The papers are grouped by theme. Each one is described in three short lines.

Theme: RAG foundations

Paper 1: Lewis et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*

- This is the paper that introduced RAG, the technique our whole project is built on.
- It joins a search component with a text-generating model so answers come from retrieved text, not only from the model's memory.
- We cite it to explain the basic pipeline: retrieve relevant text first, then generate the answer.

Theme: Citations and attribution

Paper 2: Gao et al. (2023). *Enabling Large Language Models to Generate Text with Citations (ALCE)*

- This paper is about making AI answers verifiable by attaching citations to them.
- The authors built a benchmark and measures to check how well answers are backed by their cited sources.
- It supports our citation feature and gives us ideas for measuring citation quality.

Theme: Regulatory and banking QA

Paper 3: Gokhan et al. (RIRAG) (2024). *RIRAG: Regulatory Information Retrieval and Answer Generation*

- RIRAG builds a question answering setup for financial regulations: find the right passages, then write the answer.
- Its main contribution is the ObliQA dataset plus a baseline system.
- It is the closest match to our domain and pipeline.

Paper 4: Hybrid regulatory IR paper (2025). *A Hybrid Approach to Information Retrieval and Answer Generation for Regulatory Texts*

- This paper improves the retrieval step for regulatory documents.
- It mixes keyword search with meaning-based search, then generates answers from what was found.
- It tells us that combining both search types is a useful improvement for our retrieval stage.

Paper 5: Licari et al. (2024). *A Novel Multi-Step Prompt Approach for LLM-based Q&As on Banking Supervisory Regulation*

- This paper uses LLMs to answer questions on banking supervisory rules in Europe.
- It uses step-by-step prompting with RAG and an automatic evaluator.
- It confirms that banking QA is a real need and that answers must stay checkable by humans.

Theme: Banking and financial RAG systems

Paper 6: FinRAG-12B (2026). *FinRAG-12B: a production-validated recipe for grounded question answering in banking*

- This work builds a banking-specific RAG model that stays grounded in the supplied documents.
- A key rule is that the model must refuse to answer when the information is missing.
- It supports our plan to add an 'answer not found' behaviour.

Paper 7: FinSage (2025). *FinSage: A Multi-aspect RAG System for Financial Filings Question Answering*

- FinSage is a RAG system for answering questions about financial filings.

- It uses several retrieval aspects together and was tested on expert questions and a public benchmark.
- It shows that RAG works in real financial settings, but its design is heavier than what a student project needs.

Theme: Chunking and retrieval quality

Paper 8: Jimeno Yepes et al. (2024). *Financial Report Chunking for Effective Retrieval Augmented Generation*

- This paper studies how to cut financial documents into pieces before indexing them.
- It shows that splitting by document structure works better than splitting by fixed length.
- It guides our chunking step, where each chunk should keep its page number and section.

3. Research Gap: What Nobody Has Done

After reviewing the eight papers above, these are the gaps we found. They are based on the papers we reviewed, not on every paper in the field.

- **Customer-facing bank policy documents.** The banking papers focus on regulations, supervisory rules or company filings used by compliance staff. None of the eight targets everyday bank policy documents such as loan terms, interest rates, account rules and transaction limits.
- **Citations as a central feature in banking.** Citation research (ALCE) uses general-knowledge questions, and the banking papers focus on retrieval and answer quality. None of them makes page-level citations the main feature for banking users.
- **Simple, reproducible systems.** FinRAG-12B and FinSage depend on proprietary data, model training or complex designs. Few works show a lightweight pipeline built with open tools that a small team can reproduce.
- **Indian banking context.** The datasets and documents in these papers come from the US, Europe and the UAE. None of the eight uses Indian bank policy documents.

4. How Our Project Fills This Gap

Our project is a citation-grounded RAG assistant for banking policy documents. It combines ideas from the reviewed papers in one simple, usable system.

Gap	How our project fills it
Customer-facing bank policy documents	Works on bank policy PDFs (loans, interest rates, accounts, eligibility, transaction limits) and lets anyone ask in plain English.
Citations as a central feature	Every chunk keeps its document name and page number, so each answer is shown with its source, for example 'Savings Account Policy, page 3'. Users can verify the answer themselves.
Simple, reproducible system	Uses open and accessible tools: Python, a sentence-embedding model, a vector database, the Gemini API and a React front end. No model training is needed.
Indian banking context	Uses Indian bank and RBI policy documents as the knowledge base.
Reducing made-up answers	The model is told to answer only from retrieved text and to say 'not found in the documents' when nothing relevant is retrieved (idea supported by FinRAG-12B).
Retrieval quality	Structure-aware chunking (paper 8) and optionally hybrid keyword plus semantic search (paper 4) improve what the model sees.

Novelty statement: A citation-grounded RAG assistant for Indian banking policy documents that gives page-level, verifiable answers using a simple, reproducible pipeline.

5. References

[1] Lewis et al., *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*, 2020. https://arxiv.org/abs/2005.11401

[2] Gao et al., *Enabling Large Language Models to Generate Text with Citations (ALCE)*, 2023. https://arxiv.org/abs/2305.14627

[3] Gokhan et al. (RIRAG), *RIRAG: Regulatory Information Retrieval and Answer Generation*, 2024. https://arxiv.org/abs/2409.05677

[4] Hybrid regulatory IR paper, *A Hybrid Approach to Information Retrieval and Answer Generation for Regulatory Texts*, 2025. https://arxiv.org/abs/2502.16767

[5] Licari et al., *A Novel Multi-Step Prompt Approach for LLM-based Q&As on Banking Supervisory Regulation*, 2024. https://aclanthology.org/2024.clicit-1.59.pdf

[6] FinRAG-12B, *FinRAG-12B: a production-validated recipe for grounded question answering in banking*, 2026. https://research.backbase.com/research/finrag-12b

[7] FinSage, *FinSage: A Multi-aspect RAG System for Financial Filings Question Answering*, 2025. https://arxiv.org/abs/2504.14493

[8] Jimeno Yepes et al., *Financial Report Chunking for Effective Retrieval Augmented Generation*, 2024. https://arxiv.org/abs/2402.05131